

APP DEVELOPMENT PROJECT

1. Student Details:

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About me: I am a dual-degree student pursuing the IIT Madras BS in Data Science, passionate about analytics, problem-solving, and building real-world skills.

2. Project Details:

Project Title: Placement Portal Application

Problem Statement:

Institutes need efficient systems to manage campus recruitment activities involving companies, students, and placement drives. Currently, many institutes rely on spreadsheets, emails, or manual processes, which makes it difficult to manage company approvals, track student applications, avoid duplicate registrations, and maintain placement records. You are required to build a Placement Portal Application web application that allows Admin (Institute), Company, and students to interact with the system based on their roles.

Approach:

I started the project by designing a database with SQLite3.

Using Flask and Jinja2, I developed the primary interfaces for Admins, Students, and Companies. Then I added different functions as per requirement. To ensure that all core functions are taken into consideration I verified it with Milestone Document. Changes were done till it perfectly aligned with the given functionality.

3. AI/LLM Declaration:

I used ChatGPT (GPT-5) to assist in improving database, backend routing.

The extent of AI/LLM usage is around 10–15%, limited to code suggestions.

All final implementation logic, debugging, and integration were done manually.

Component / Tool	Purpose / Role in Project	Usage %
Flask (App + API)	Backend logic, routing, request handling, sorting.	20%
SQLite	Data modeling, relationships, query	10%
HTML + Jinja2	Server-side template rendering	5%
Bootstrap / CSS	Styling and layout	3%

4. Technologies and Framework Used:

Technology / Library	Specific Purpose and Application within the Project
Python 3	Core programming language for the application backend and database interaction scripts.
Flask	Core backend web framework used for routing, request handling, session management.
SQLite	Lightweight local database for storing all student and company records.
Jinja2	Template engine for rendering dynamic HTML pages based on backend data.
CSS3, Bootstrap 5	Frontend styling, responsive design, visual design, including the layout of dashboards and the styling of tables and buttons.

5. Database Schema / ER Diagram

Tables:

admin – stores admin details (id, username, password)

students – stores student details (id, name, email, password, contact, resume, status)

companies – stores company details (id, name, email, password, website, hr_contact, status)

drives – stores placement drive details posted by companies (id, company_id, job_title, skills, experience, salary, status)

applications – stores student applications to drives (id, student_id, drive_id, status)

Relationships:

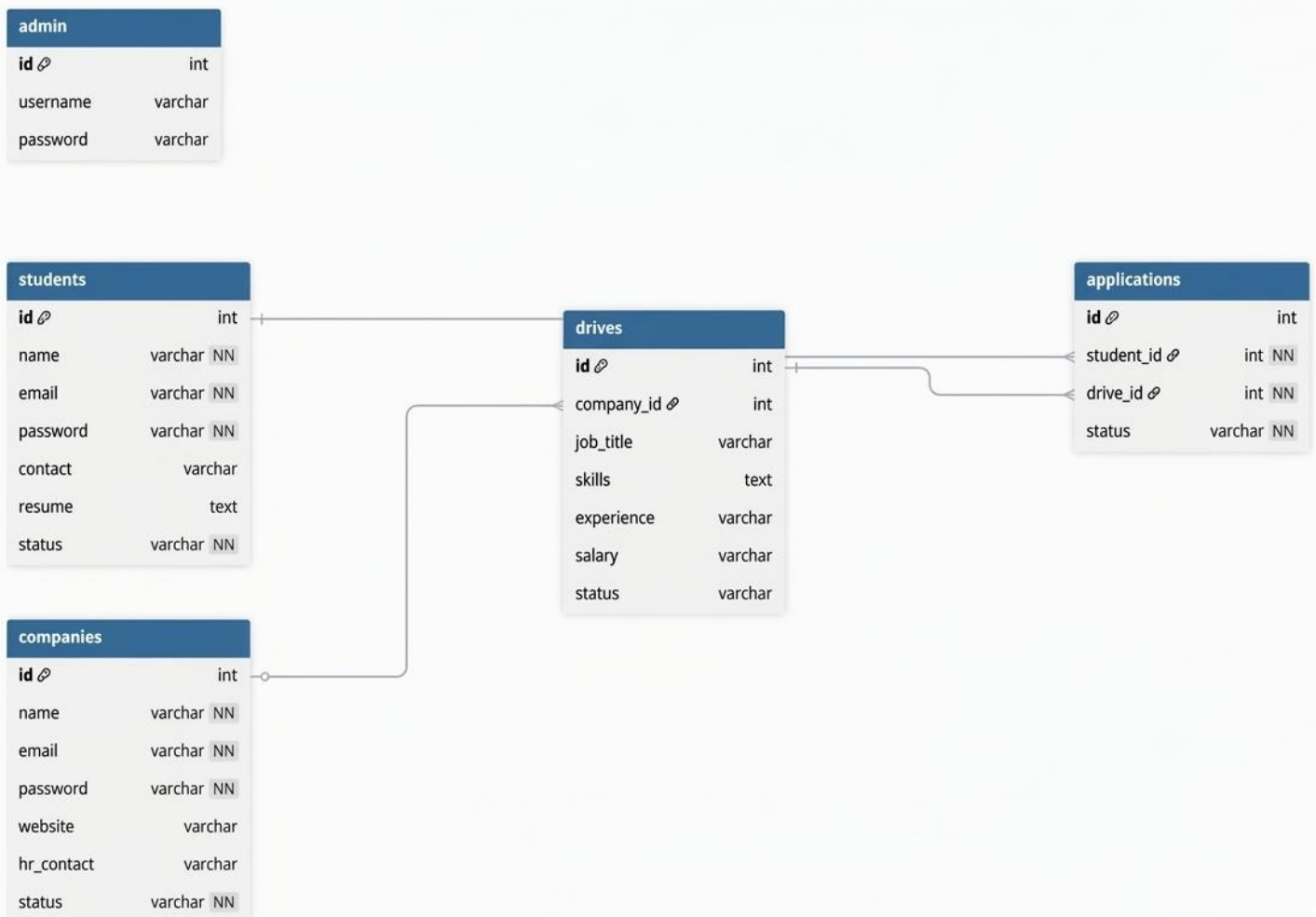
One-to-Many: companies → drives

One-to-Many: students → applications

One-to-Many: drives → applications

Many-to-Many: students ↔ drives (via applications)

No relationship: admin



6. Architecture and Features:

Architecture Overview:

app.py – main flask application entry point containing all route definitions and business logic

database.py – database models using SQLite, for database connectivity and schema initialization.

/templates – folder containing Jinja2 HTML templates organized into subfolders for admin/, company/, and student/ specific views

/static – folder containing style.css for application styling.

Implemented Features:

Admin: The Admin acts as a superuser, approving or rejecting company sign-ups and job postings to keep things organized. They have the power to manage everyone's data and can block any user.

Company: Companies can login and post job and find new talent. They review student resumes, pick the best candidates.

Student: Students create their profiles and upload resumes to find and apply for jobs. They can search job, track their applications.

7. Video Presentation:

Drive Link:

https://drive.google.com/file/d/1N3xhogXwm1yH8YQ8Tk_--q1tyg83vJ9P/view?usp=sharing